

Tide retrospective: third-moment asymptotic for the anharmonic Gibbs

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7 May 2026

Abstract

This retrospective records the smallest tide on the board: a ~ 30 -line strict-improvement of Tide 9. The third moment $\langle x^3 \rangle_t$ of the anharmonic Gibbs measure was internal to the proof of `cov_anharmonic_asymptotic` (Tide 9); this tide packages it as a separate user-facing theorem $t^2 \cdot \langle x^3 \rangle_t \rightarrow -5\alpha/(2\lambda^3)$ via the algebraic identity $\langle x^3 \rangle = \text{Cov}[x^2, x] + \langle x^2 \rangle \langle x \rangle$ combined with the existing asymptotics. Zero `sorry`, single session, no GPT-5.5 Pro consult needed (the deliberation already happened in Tide 9’s retrospective).

1 Setting

Tide 9 (anharmonic- κ_3 , 6 May) proved $t^2 \cdot \kappa_3(x, x, x) \rightarrow -\alpha/\lambda^3$ for the 1D anharmonic Gibbs measure. The proof factored through three asymptotics already in the laplace seabed:¹ the covariance, the mean, and the variance. Internal to the strategy was the third moment expansion $\langle x^3 \rangle_t = -5\alpha/(2\lambda^3 t^2) + o(t^{-2})$, which appeared as the corrected Wick coefficient (the famous “5/2 vs 3/2” correction caught by both the numerical sanity check and the GPT consult).

That third moment was never packaged as a stand-alone user-facing theorem. Tide 9’s retrospective explicitly named it as a deferred follow-up: “*With Tide 9’s moment expansions exposed, $t^2 \cdot \langle x^3 \rangle_t \rightarrow -5\alpha/(2\lambda^3)$ is a ~ 5 -line corollary of $\text{Cov}[x^2, x] = \langle x^3 \rangle - \langle x^2 \rangle \langle x \rangle$ plus existing asymptotics. Worth packaging as `thirdMoment_anharmonic_asymptotic` for downstream consistency.*”

This tide closes that follow-up, exactly as Tide 9 anticipated.

2 The thing formalised

For the 1D anharmonic potential $L(x) = (\lambda/2)x^2 + (\alpha/6)x^3 + (\gamma/24)x^4$ with $\lambda > 0$, $\gamma > 0$, and the discriminant condition $\alpha^2 < 3\lambda\gamma$,

$$t^2 \cdot \langle x^3 \rangle_t \longrightarrow -\frac{5\alpha}{2\lambda^3} \quad \text{as } t \rightarrow \infty.$$

The Lean signature, in `Laplace/OneD/AnharmonicKappa3.lean`²:

¹Specifically, in `Laplace/OneD/IntegralRemainder.lean`: `cov_anharmonic_asymptotic` (the covariance), `mean_anharmonic_asymptotic` ($\langle x \rangle$), and `cov_self_anharmonic_asymptotic` (the variance).

²The new theorem sits at the bottom of that file; the existing κ_3 infrastructure remains unchanged. Permalink: the file at the merged commit on `main`.

```

theorem thirdMoment_anharmonic_asymptotic
  {lam alpha gamma : ℝ}
  (hlam : 0 < lam) (hgamma : 0 < gamma)
  (hdisc : alpha ^ 2 < 3 * lam * gamma) :
  Filter.Tendsto
    (fun t : ℝ => t ^ 2 * Laplace.gibbsExpectation
      (anharmonicPotential lam alpha gamma) t (fun x : ℝ => x ^ 3))
    Filter.atTop
    (nhds (-(5 * alpha) / (2 * lam ^ 3)))

```

The hypotheses are inherited verbatim from Tide 9.

3 Proof strategy

The proof is short enough to write out completely in prose:

By the definition of `gibbsCov`,

$$\text{Cov}_t[x^2, x] = \langle x^3 \rangle_t - \langle x^2 \rangle_t \cdot \langle x \rangle_t,$$

so

$$\langle x^3 \rangle_t = \text{Cov}_t[x^2, x] + \langle x^2 \rangle_t \cdot \langle x \rangle_t.$$

Multiplying by t^2 :

$$t^2 \cdot \langle x^3 \rangle_t = t^2 \cdot \text{Cov}_t[x^2, x] + (t \cdot \langle x^2 \rangle_t) \cdot (t \cdot \langle x \rangle_t).$$

Each factor on the right has a known limit from Tide 9's bag of asymptotics: $-2\alpha/\lambda^3$ for the covariance, $1/\lambda$ for $t \cdot \langle x^2 \rangle$, and $-\alpha/(2\lambda^2)$ for $t \cdot \langle x \rangle$. By `Filter.Tendsto.add` + `Filter.Tendsto.mul`:

$$t^2 \cdot \langle x^3 \rangle_t \longrightarrow -\frac{2\alpha}{\lambda^3} + \frac{1}{\lambda} \cdot \left(-\frac{\alpha}{2\lambda^2}\right) = -\frac{5\alpha}{2\lambda^3}.$$

The terminal arithmetic is one `field.simp + ring`.

The Lean structure follows the proof verbatim: 1. Pull in the three asymptotic hypotheses (`hCov`, `hM2`, `hM1`). 2. Build the sum-of-products via `hCov.add (hM2.mul hM1)` — this gives the limit value $-2\alpha/\lambda^3 + (1/\lambda) \cdot (-\alpha/(2\lambda^2))$. 3. Rewrite the limit value to $-5\alpha/(2\lambda^3)$ via a `have h_lim_eq + field.simp ; ring`. 4. Bridge the function: the integrand on the LHS $t^2 \cdot \langle x^3 \rangle$ equals the integrand on the RHS $t^2 \cdot \text{Cov}[x^2, x] + (t \cdot \langle x^2 \rangle)(t \cdot \langle x \rangle)$ after unfolding `gibbsCov` and a small `x^2 * x = x^3` function-extensionality nudge. 5. `Tendsto.congr` closes the goal.

4 Roadblocks and resolutions

There were no roadblocks. The proof followed the deliberation plan verbatim, and the build was clean on first pass after one minor lint warning (`show` vs `change`, which the linter handles).

The only technical detail worth recording: Tide 9's `cov_anharmonic_asymptotic` is stated with the derived limit value $-2\alpha/\lambda^3$ (not in the $-2\alpha/\lambda^3 + (1/\lambda) \cdot (-\alpha/(2\lambda^2))$ form), so the limit-value rewrite step had to combine the two sub-asymptotics' limits into the headline limit value via `ring arithmetic`. The `nhds (a + b)-vs-nhds c` pattern that recurred in Tides 9 and 10 reappeared here: build the sum, name the equality, rewrite. Three minutes of plumbing.

5 What was Mathlib, what was new

Mathlib provided. The standard `Tendsto`-algebra glue (`add`, `mul`, `congr'`, `filter_upwards`). Nothing exotic.

Laplace provided. Three asymptotics from Tide 9 / the original anharmonic stack: covariance, second moment, and mean.³

What was new. One theorem (`thirdMoment_anharmonic_asymptotic`, ~30 lines) and one limit-value rewrite. No new helper lemmas were introduced. The mathematical content was already in the Tide 9 retrospective; this tide is pure packaging.

6 Lessons

- **Strict-improvement follow-ups left in retrospectives are load-bearing.** Tide 9’s retrospective named exactly this corollary, with the closed-form value, with the proof outline, with the line-count estimate. Today’s tide consumed that named follow-up directly. The `tide-pick` skill’s emphasis on harvesting retrospectives’ Follow-up sections paid off: this tide cost almost nothing because the deliberation had already happened.
- **Skip the GPT consult for genuine packaging tides.** For a follow-up that’s already named in a prior retrospective with the explicit closed form and proof outline, a fresh GPT-5.5 Pro consult is overhead. The deliberation lives in the prior retrospective; the current tide’s job is execution. Worth landing in the tide skill: *a “packaging” modifier where Step 2 is reduced to a single-candidate log entry and direct Step 3 hand-off, when the candidate is a named follow-up of a closed retrospective.*
- **The same plumbing pattern recurs.** The three Tides 9, 10, and now this G3 all share a structure: sum/product of two-three existing `Tendsto` facts plus a limit-value `ring` rewrite plus a `Tendsto.congr'` bridge. Worth abstracting once the pattern fires a fourth time (the `gibbsCov_symm` + covariance-algebra refactor in I2 from the survey is exactly this).

7 Follow-ups

- *Affine-observable third moment.* Generalise to $t^2 \cdot \langle (ax+c)^3 \rangle_t \rightarrow \dots$ via multilinearity and the existing $\langle x \rangle, \langle x^2 \rangle$ asymptotics. ~30 extra lines. Pairs with G2 (affine-observable anharmonic κ_3).
- *Higher-moment asymptotics.* The same template lifts to $\langle x^4 \rangle, \langle x^5 \rangle$ and beyond — but the sub-asymptotics for those moments are not yet exposed in `laplace` and the leading orders shift. Sequenced: would build atop a “`thirdMoment` via the rescaling identity” refactor first.
- *Update the survey.* Mark G3 closed.

³Concretely `cov.anharmonic_asymptotic`, `secondMoment_anharmonic_asymptotic`, and `mean_anharmonic_asymptotic`.

Acknowledgements

This tide skipped the deliberation step entirely. The deliberation was inherited verbatim from Tide 9’s retrospective Follow-ups section, which named the closed form, the proof outline, and the line-count estimate. GPT-5.5 Pro contributed nothing to this tide directly — its load-bearing contribution was at Tide 9’s deliberation, where it caught the Wick-coefficient error that exposed the correct $5\alpha/2$ term.